Labaratory 05 Sequence Models
RNN, LSTM, and GRU

Theory Recap

from document representation to sequence

In lab 04 we ~~have~~ had

Document —> fixed size vector —> ~~clf~~ classifier

example with Average Word2Vec

$$\vec{x} = \frac{1}{T}\sum_{t=1}^{T}\vec{w_t}$$

In sequence models

$$\begin{tikzcd}
w_1 \arrow[r] & w_2 \arrow[r] \arrow[d] & w_3 \arrow[r, phantom, "\cdots"] \arrow[d] & \arrow[r] & w_t \arrow[d] \\
& h_1 & h_2 & \text{--} & h_t
\end{tikzcd}$$

each token will ~~proess~~ to be processed by ordreration

Hiddenstate

$$h_t = f(x_t ; h_{t-1})$$

so

h_t have informations previous token information.

Unimodal RNN

$$h_t = \tanh(\underline{W_{xh}} \cdot \underline{x_t} + \underline{W_{hh}}\, \underline{h_{t-1}} + \underline{b_h})$$

Embedding of token at t

bias

Previous token state

Input to
hidden
weight
(learnable)

Recurrent weight
(learnable)

output classification was computed by

$$z = W_{hh} h_T + b_y$$

and then

$$P(y|x) = \text{softmax}(z)$$

With binary classification task we can use
sigmoid function

Calculation

Bài 1

$x_t = \begin{bmatrix}1\\0\end{bmatrix}$ $h_{t-1} = \begin{bmatrix}0{,}5\\0{,}2\end{bmatrix}$ $w_{xh} = \begin{bmatrix}0{,}5 & 0{,}2\\0{,}1 & 0{,}4\end{bmatrix}$

$w_{hh} = \begin{bmatrix}0{,}3 & 0{,}1\\0{,}2 & 0{,}5\end{bmatrix}$ $bias = \begin{bmatrix}0\\0\end{bmatrix}$

$a_t = w_{xh} x_t + w_{hh} \cdot h_{t-1}$

$= \begin{bmatrix}0{,}5 & 0{,}2\\0{,}1 & 0{,}4\end{bmatrix}\begin{bmatrix}1\\0\end{bmatrix} + \begin{bmatrix}0{,}3 & 0{,}1\\0{,}2 & 0{,}5\end{bmatrix}\begin{bmatrix}0{,}5\\0{,}2\end{bmatrix}$

$= \begin{bmatrix}0{,}5\\0{,}1\end{bmatrix} + \begin{bmatrix}0{,}17\\0{,}2\end{bmatrix} = \begin{bmatrix}0{,}67\\0{,}3\end{bmatrix}$

$\Rightarrow h_t = \tanh(a_t) = \begin{bmatrix}\tanh(0{,}67)\\\tanh(0{,}3)\end{bmatrix} =$ $\cancel{e^{0{,}67} - e^{0{,}}}$

$\tanh(x) = \dfrac{e^{x} - e^{-x}}{e^{x} + e^{-x}}$

Bài 2 Information

"I did not like this movie"

$h_1 = f(h_0 ; \text{"I"}) \rightarrow h_2 = f(h_1 ; \text{"did"})$

$h_3 = f(h_2 \text{ "not"}) \dots \quad h_6 = (h_5 \text{ "movie"})$

Bài 3

S_1 = "dog bites man"

S_2 = "man bites dog"

* gọi ~~$\vec{v_1}$ = dog~~

ve

"dog" ~~will be~~ could be presented by $\vec{v_1}$

"bites" ______ $\vec{v_2}$

"man" ______ v_3

↳ $\vec{v_{s1}} = \frac{1}{3}(\vec{v_1} + \vec{v_2} + \vec{v_3})$

$\vec{v_{s2}} = \frac{1}{3}(\vec{v_1} + \vec{v_2} + \vec{v_3})$

=> $\vec{v_{s1}} = \vec{v_{s2}}$

↳ word2vec không phân biệt được

~~vo~~ with RNN models

~~$v_{s_1} = h_3 = f(h_2$ man~~

$h_3' = f(f(f(\overset{h_0}{\cancel{dog}}\ \overset{dog}{\cancel{h_0}})\ \&\ bites)\ man)$

$h_3 = f(f(f(h_0\ man)\ bites)\ dog)$

$h_3 \neq h_3'$ => câu